Cross-validated Targeted Maximum Likelihood Estimation (CV-TMLE)

Double machine learning for added robustness

Alan Hubbard

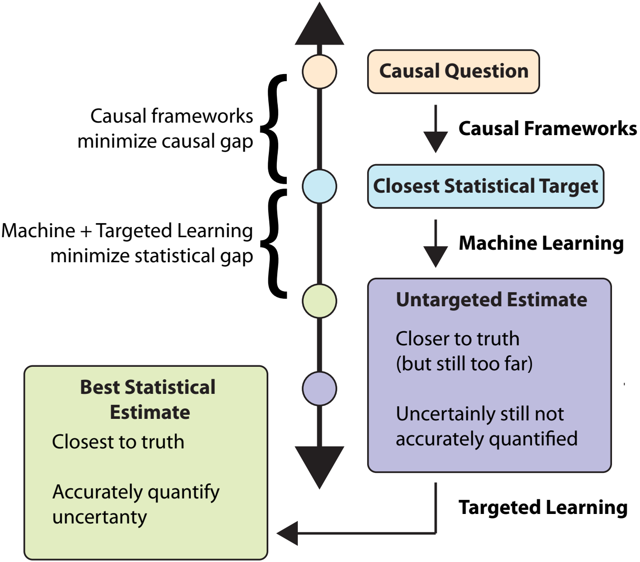
Division of Biostatistics
University of California at Berkeley

October 22 2020

Done in Collaboration with Haodong Li, Andrew Mertens, Jack Colford, Jeremy Coyle, Mark van der Laan and ki software team at UC Berkeley and the ki program at Gates
This research was financially supported by a global development grant (OPP1165144) from the Bill Melinda Gates Foundation to the University of California, Berkeley, CA, USA

Outline

Targeted Learning fills a much needed gap in machine learning and causal inference



Roadmap for Targeted Learning

1. Describe observed data
2. Specify statistical model
3. Define statistical query (e.g., using causal roadmap)
4. Construct estimator
5. Obtain inference

Roadmap for Targeted Learning

STEP 1:
DESCRIBE
OBSERVED DATA

STEP 2:
SPECIFY
STATISTICAL MODEL

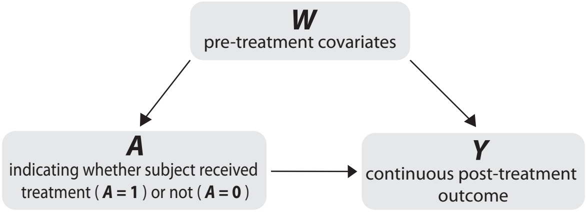
STEP 3:
DEFINE
STATISTICAL QUERY

STEP 4:
CONSTRUCT
ESTIMATOR

STEP 5:
OBTAIN INFERENCE

$n = 100$ subjects were sampled independently from each other and from the same population distribution P_0

For each subject, pre-treatment covariates (W), treatment (A), and outcome (Y) vectors were measured



Roadmap for Targeted Learning

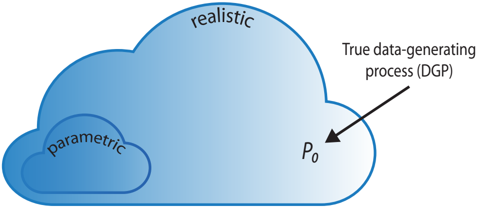
STEP 1:
DESCRIBE
OBSERVED DATA

STEP 2:
SPECIFY
STATISTICAL MODEL

STEP 3:
DEFINE
STATISTICAL QUERY

STEP 4:
CONSTRUCT
ESTIMATOR

STEP 5:
OBTAIN INFERENCE



Standard Approach	Targeted Learning
Parametric statistical model	Realistic semiparametric or nonparametric statistical model
Does not contain P_0, the DGP (i.e., misspecified model)	Defined to ensure P_0 is contained in model

Roadmap for Targeted Learning

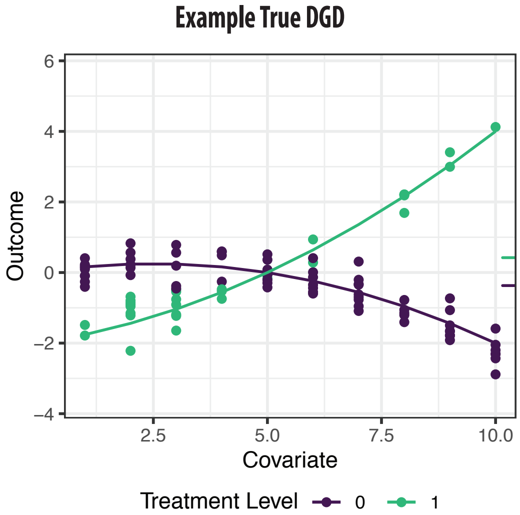
STEP 1:
DESCRIBE
OBSERVED DATA

STEP 2:
SPECIFY
STATISTICAL MODEL

STEP 3:
DEFINE
STATISTICAL QUERY

STEP 4:
CONSTRUCT
ESTIMATOR

STEP 5:
OBTAIN INFERENCE



Roadmap for Targeted Learning

STEP 1:
DESCRIBE
OBSERVED DATA

STEP 2:
SPECIFY
STATISTICAL MODEL

STEP 3:
DEFINE
STATISTICAL QUERY

STEP 4:
CONSTRUCT
ESTIMATOR

STEP 5:
OBTAIN INFERENCE

What is the average difference in outcomes between treatment groups when adjusting for covariates?

$$\Psi(P_0) = E_0\big(E_0[Y|A=1,W] - E_0[Y|A=0,W]\big)$$

Ψ is a function that takes as input P_0
and outputs the answer to the question of interest

The ***assumption of positivity*** is requied to estimate of this quantity from the data. That is, it must be possible to observe both levels of treatment for all strata of *W*.

Additional assumptions are required to interpret this estimand as causal

Roadmap for Targeted Learning

STEP 1:
DESCRIBE
OBSERVED DATA

STEP 2:
SPECIFY
STATISTICAL MODEL

STEP 3:
DEFINE
STATISTICAL QUERY

STEP 4:
CONSTRUCT
ESTIMATOR

STEP 5:
OBTAIN INFERENCE

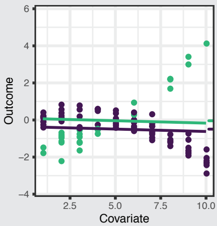
Standard Approach

Generalized Linear Model (GLM) to estimate

$$\mathbf{Y} = \beta_0 + \beta_1 \mathbf{A} + \beta_2 \mathbf{W} + \epsilon$$

Estimated coefficients are biased

Cannot detect heterogeneity in treatment effect

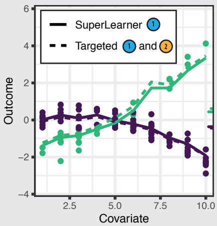


Targeted Learning

TMLE implements a two-step procedure

1. initial estimation of $E_0[Y|A, W]$ with super (machine) learning
2. targeting towards optimal bias-variance trade-off for $\Psi(P_0)$

TMLE estimates are unbiased and doubly robust



Roadmap for Targeted Learning

STEP 1:
DESCRIBE
OBSERVED DATA

STEP 2:
SPECIFY
STATISTICAL MODEL

STEP 3:
DEFINE
STATISTICAL QUERY

STEP 4:
CONSTRUCT
ESTIMATOR

STEP 5:
OBTAIN
INFERENCE

Standard Approach

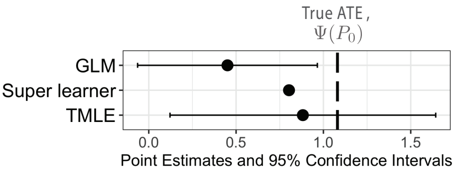
Inference (such as *p*-value and confidence interval) assumes parametric model is true

Inference is misleading and erroneous

Targeted Learning

Targeting (step 2) improves estimate and makes inference possible

Trustworthy inference obtained with efficient influence function

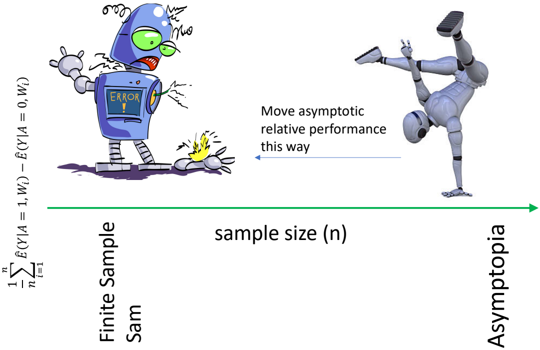


What do you want in a statistical machine?

- Takes in inputs all constraints on the likelihood and causal model and the target quantity of interest.
- Optimizes the estimation of target estimand with fidelity to the stated constraints.
  - ✓ constraints on data-generating distribution
  - ✓ causal model where known
  - ✓ etc.
- Returns efficient estimates and inference or,
- tells you the information not available to estimate quantity.
- Uses data-adaptive methods that account for sample size, smoothness of data-generating distribution, etc.

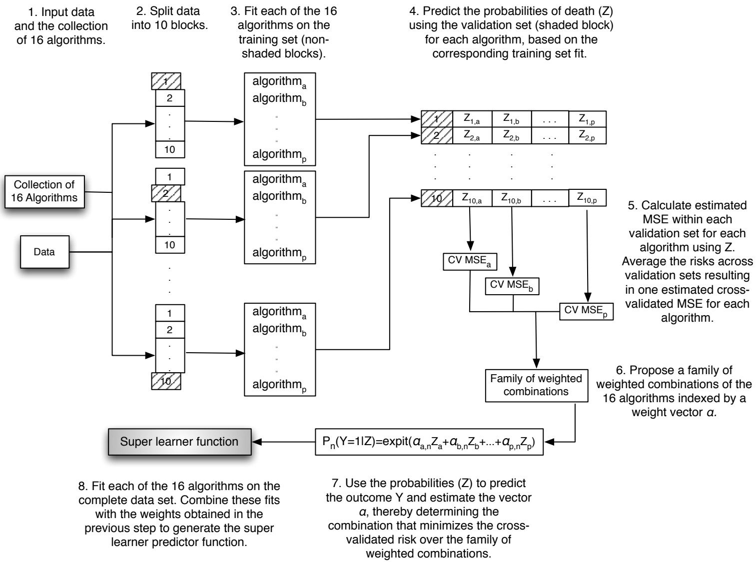
How theory can help

- Theory can predict asymptotic performance and help to define the most efficient estimator in the most realistic model.
- Theory not always available for optimizing finite sample performance, but can give insights into improving performance.
- Thus, simulations, particularly "realistic" simulations can provide the best evidence of how well the proposed machine performs in practice.



Outline

Super Learner - a talented ensemble



Super Learner: need to make the Oracle smart by using large library

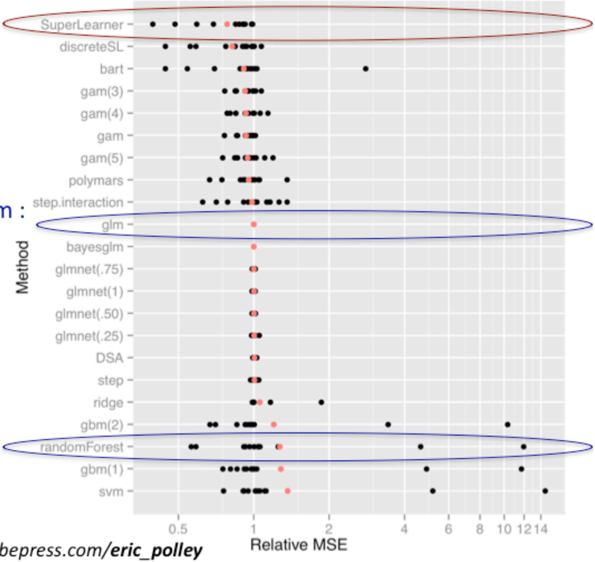
- The super learner improves asymptotically on the discrete super learner by **working with a larger library.**
- Asymptotic results prove that in realistic scenarios (where none of the algorithms are a correctly specified parametric model), the discrete super learner **performs asymptotically as well as the *oracle***, which we define as the best estimator given the algorithms in the collection of algorithms.
- Consequently, super learner performs asymptotically as well as the best choice among the family of weighted combinations of estimators.
- Thus, *by adding more competitors, we only improve the performance of the super learner.*

Works well in theory and practice

Super Learner- Best weighted combination of algorithms for a given prediction problem

Example algorithm : Linear Main Term Regression

Example algorithm: Random Forest



Technical Report: *works.bepress.com/**eric_polley***

Substitution Estimator for Average Treatment Effect (ATE), $\Psi(P_X) = \mathbb{E}(Y_1 - Y_0)$

- The Estimand:
  - ✓ Data is: $O = (W, A, Y) \sim P_0 \in \mathcal{M}^{NP}$ (nonparametric model)
  - ✓ Under assumptions discussed earlier, $\Psi(P_X) = \Psi(P_0) = \Psi(Q_0)$, where Q_0 represents both the distribution of $Y \mid W, A$ and distribution of W.

$$\Psi(Q_0) = E_{W,0}\{E_0(Y \mid A = 1, W) - E_0(Y \mid A = 0, W)\}$$

  - ✓ Let $Q_0(A, W) \equiv E_0(Y \mid A, W)$ and $Q_{0,W}(w) = P_0(W = w)$, then

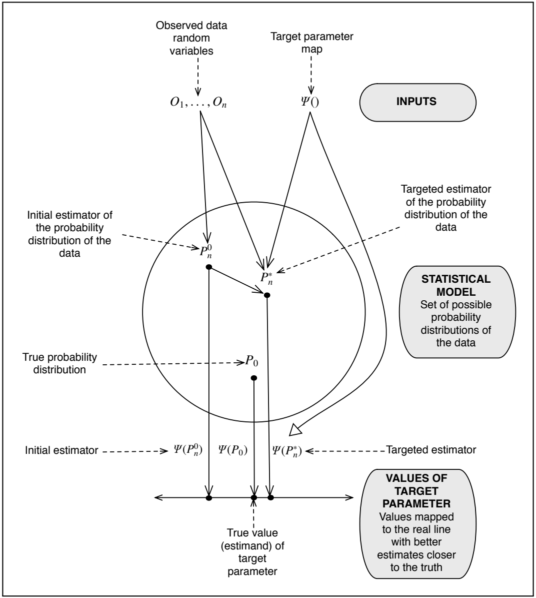
$$\Psi(Q_0) = \sum_w \{Q_0(1, w) - Q_0(0, w)\} Q_{0,W}(w)$$

- The Substitution Estimator

$$\Psi(Q_n) = \frac{1}{n} \sum_{i=1}^{n} \{Q_n(1, W_i) - Q_n(0, W_i)\}$$

  - ✓ $Q_{n,W}(W_i) = 1/n$ (the empirical) and $Q_n(A, W)$ is a regression (or machine learning) fit of Y on (A, W).

TMLE Algorithm



Targeted MLE

1. Identify the parametric model for fluctuating initial $\hat{P}$
   – Small "fluctuation" $\rightarrow$ maximum change in **target**.
2. Given strategy, identify optimum amount of fluctuation by MLE.
3. Apply optimal fluctuation to $\hat{P}$ $\rightarrow$ **1st-step targeted maximum likelihood estimator**.
4. Repeat until the incremental "fluctuation" is zero
   – Some important cases: 1 step to convergence.
5. Final probability distribution solves efficient influence curve equation

$\rightarrow$ **T-MLE is double robust & locally efficient**

TMLE for the Average Causal Effect

- TMLE for the ATE is a two-step procedure, that updates the original fit of Q_n with a clever covariate, treating the initial fit of Q as an offset.
- The first step is to generate an initial estimator of P_n^0 of P with super learner (van der Laan et al. (2007)) to the entire data (P_n).
- We fluctuate this initial estimator with a logistic regression:

$$\text{logit} P_n^0(\epsilon)(Y = 1 \mid A, W) = \text{logit} P_n^0(Y = 1 \mid A, W) + \epsilon h$$

where

$$H_n^*(A, W) \equiv \left(\frac{I(A = 1)}{g_n(1 \mid W)} - \frac{I(A = 0)}{g_n(0 \mid W)} \right).$$

- Update the estimate $\bar{Q}_n^0$ into a new estimate $\bar{Q}_n^1$ of the true regression function $\bar{Q}_0$:

$$\text{logit } \bar{Q}_n^1(A, W) = \text{logit } \bar{Q}_n^0(A, W) + \epsilon_n H_n^*(A, W).$$

- This parametric working model incorporated information from g_n, through $H_n^*(A, W)$, into an updated regression, $logit(\bar{Q}_\epsilon(P_n)) = logit(\hat{\bar{Q}}(P_n)) + \epsilon_n^*(A, W)$
- The TMLE is given by:

$$\Psi(Q_n^*) = \frac{1}{n} \sum_{i=1}^{n} \{\bar{Q}_n^1(1, W_i) - \bar{Q}_n^1(0, W_i)\}.$$

Conditions for asymptotic linearity for TMLE

- Let $\Psi(p_0)$ be the parameter of interest and $\Psi(p_n^*)$ be the TMLE estimator.
- One can write the linearization of TMLE estimator as as

$$\Psi(p_n^*) - \Psi(p_0) = (P_n^* - P_0)D(p_n^*) + R(n).$$

- Applying empirical process theory to show asymptotic linearity requires that $D(p_n^*)$ falls in a P_0-Donsker class with probability tending to 1.
- Without explanation, this requires that the estimators Q_n and g_n for $Q_0(A, W) \equiv E_0(Y \mid A = a, W)$ and $g_0(W) \equiv P_0(A = 1 \mid W)$ are not "too" adaptive.

Proof is in the pudding

Authors	Title	year	Pro/Con
Rose and van der Laan	Simple Optimal Weighting of Cases and Controls in Case-Control Studies	2008	Pro
Moore and van der Laan	Covariate adjustment in randomized trials with binary outcomes: targeted maximum likelihood estimation	2009	Pro
Gruber and van der Laan	An application of collaborative targeted maximum likelihood estimation in causal inference and genomics	2010	Pro
Stitelman and van der Laan	Collaborative Targeted Maximum Likelihood for Time to Event Data	2010	Pro
Porter, et al.	The relative performance of targeted maximum likelihood estimators	2011	Pro
Wang, et al.	Finding Quantitative Trait Loci Genes with Collaborative Targeted Maximum Likelihood Learning	2011	Pro
Muñoz and van der Laan	Population Intervention Causal Effects Based on Stochastic Interventions	2011	Neutral
	Diagnosing and responding to violations in the positivity assumption	2012	Pro
van der Laan and Gruber	Targeted Minimum Loss Based Estimation of Causal Effects of Multiple Time Point Interventions	2012	Pro
Gruber and van der Laan	An Application of Targeted Maximum Likelihood Estimation to the Meta-Analysis of Safety Data	2013	Neutral
Lendle, et al.	Targeted maximum likelihood estimation in safety analysis	2013	Pro
Díaz and van der Laan	Targeted Data Adaptive Estimation of the Causal Dose Response Curve	2013	Pro
Schnitzer, et al.	time-dependent treatment effects under density misspecification	2013	Neutral
Kreif, et al.	Misspecification: a comparison of targeted maximum likelihood estimation with bias-corrected matching.	2014	Pro
Schnitzer, et al.	Effect of breastfeeding on gastrointestinal infection in infants: A targeted maximum likelihood approach for clustered longitudinal data	2014	Pro
Pang, et al.	Effect Estimation in Point-Exposure Studies with Binary Outcomes and High-Dimensional Covariate Data--A Comparison of Targeted Maximum Likelihood Estimation and Inverse Probability of Treatment Weighting	2016	Pro

Authors	Title	year	Pro/Con
Pang, et al.	Effect Estimation in Point-Exposure Studies with Binary Outcomes and High-Dimensional Covariate Data--A Comparison of Targeted Maximum Likelihood Estimation and Inverse Probability of Treatment Weighting	2016	Pro
Schnitzer, et al.	Variable Selection for Confounder Control, Flexible Modeling and Collaborative Targeted Minimum Loss-Based Estimation in Causal Inference	2016	Pro
Zheng, et al.	Doubly Robust and Efficient Estimation of Marginal Structural Models for the Hazard Function	2016	Pro
Schnitzer, et al.	Double robust and efficient estimation of a prognostic model for events in the presence of dependent censoring	2016	Pro
Schuler and Rose	Targeted maximum likelihood estimation for causal inference in observational studies	2017	Pro
Gruber and van der Laan	Collaborative Targeted Maximum Likelihood Estimation to Assess Causal	2018	Pro
Pirracchio, et al.	Collaborative targeted maximum likelihood estimation for variable importance measure: Illustration for functional outcome prediction in mild traumatic brain injuries	2018	Pro
Luque-Fernandez, et al.	Targeted maximum likelihood estimation for a binary treatment: A tutorial	2018	Pro
Levy, et al.	A Fundamental Measure of Treatment Effect Heterogeneity	2018	Pro
Ju, et al.	dimensional data	2019	Pro
Ju, et al.	On adaptive propensity score truncation in causal inference	2019	Pro
Bahamyirou, et al.	using machine learning methods with doubly robust causal estimators	2019	Pro
Wei, et al.	A Data-Adaptive Targeted Learning Approach of Evaluating Viscoelastic Assay Driven Trauma Treatment Protocols	2019	Pro
Rudolph, et al.	Complier Stochastic Direct Effects: Identification and Robust Estimation	2019	Pro
Chatton, et al.	targeted maximum likelihood estimator for causal inference with different covariates sets: a comparative simulation study	2020	Con
Talbot an Beaudoin	A generalized double robust Bayesian model averaging approach to causal effect estimation with application to the Study of Osteoporotic Fractures	2020	Con

The CV-TMLE Estimator

Simple average over validation samples

Target parameter mapping

$$\psi_n^* = E_{B_n} \Psi(\hat{Q}_{\epsilon_n}(P^0_{B_n,n})).$$

CV-TMLE Estimator

$n \; x \; V$
matrix of indicators of validation sample

The fit of the data-generating model ($\hat{Q}_{\epsilon_n}$) based upon training sample-based fits ($P^0_{B_n n}$) and pooled TMLE update (ϵ_n).

The CV-TMLE Estimator for the ATE

- Recall, the TMLE is given by:

$$\Psi(Q_n^*) = \frac{1}{n}\sum_{i=1}^{n}\{\bar{Q}_n^1(1, W_i) - \bar{Q}_n^1(0, W_i)\}.$$

- The general one-step CV estimator

$$\psi_n^* = E_{B_n}\Psi(\hat{Q}_{\epsilon_n}(P^0_{B_n,n}))$$

- CV-TMLE for ATE

$$\psi_n^* = \frac{1}{V}\sum_{v=1}^{V}\frac{1}{n_v}\sum_{i=1}^{n} I(B_n(i,v)=1)\{\hat{\bar{Q}}_{\epsilon_n}(P^0_{B_n(\cdot,v)})(1, W_i) - \hat{\bar{Q}}_{\epsilon_n}(P^0_{B_n(\cdot,v)})(0, W_i)\}.$$

where $n_v = \sum_{i=1}^{n} B_n(i,v)$

How does this change the model under which the estimator is asymptotically linear?

- Taking some short cuts on notation, one can represent the linearization of the CV-TMLE estimator as:

$$\psi_n^* - \psi_0 = E_{B_n}(P^1_{B_n,n} - P_0)D(\hat{Q}_{B_n}(P_n), \hat{g}(P^0_{B_n,n})) + R(n)$$

- Involves a cross-validated empirical process term applied to the efficient influence curve and a second-order remainder term.
- The cross-validated empirical process term implies, for each sample split, a empirical mean over a validation sample of an estimated efficient influence curve that is (largely) estimated based on the training sample.
- It thus suggests one can establish a CLT for this cross-validated empirical process term without having to enforce restrictive entropy conditions (and thereby limit the adaptiveness of the initial estimators).
- Shown formally in Zheng and Van Der Laan (2010) (see also chapter 27 in van der Laan and Rose (2011)).

Outline

1. Background
2. Theory Lite
   - Initial fit
   - TMLE
3. **Methods**
4. Results
   - Detailed look at one Study - iLiNS Zinc
5. Changing the parameter of interest a bit
6. Wrap-up

Realistic simulation study

Study ID	Name	Location	Sample Size
1	SAS Food Supplementation	India	418
2	WASH Benefits Bangladesh	Bangladesh	4863
3	WASH Benefits Kenya	Kenya	7399
4	iLiNS-DYAD-M	Malawi	1204
5	Tanzania Child 2	Tanzania	2396
6	iLiNS-Zinc	Burkina Faso	3265
7	iLiNS-DOSE	Malawi	1931
8	LCNI-5	Malawi	840
9	JiVitA-3	Bangladesh	27275
10	JiVitA-4	Bangladesh	5443

- Data from ten nutrition interventions from various regions of Africa and South Asia. In all studies, the measured outcome was a height-to-age z-score for children from birth to 24 months (Mertens et al. (2020)).
- Interventions were all nutrition-based multi-arm interventions.
- For the purposes of this analysis, multi-level interventions were simplified to a binary treatment variable.
- Different covariates were measured among these studies, but there was significant overlap.
- Each study used a form of machine-learning (super-learning) based bootstrap (Petersen et al. (2012)) to construct realistic simulations.

Simulation Structure

For each study we go through the following process

Estimate the parts of the distribution need for simulation

- *P(W)* - use empirical
- *P(A=1|W)* - flexible SL
- *E(Y|A,W)* - flexible SL
- *var(Y|A,W)*, estimated as CV-residual error

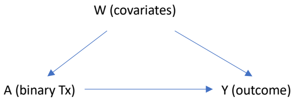
Simulate data from the fit

- *W~P_n(W)*
- $A \sim \hat{P}_{SL}(A|W)$
- $Y \sim N(\hat{E}_{SL}(Y|A,W), \widehat{var}_{CV})$

Estimate ATE and get CI for competing methods

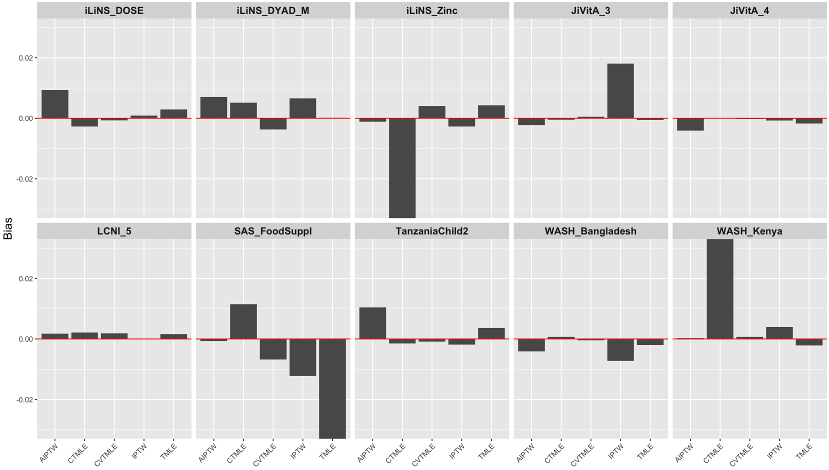
- IPTW
- AIPTW
- TMLE no CV
- CV-TMLE
- C-TMLE

Store and repeat 1000 times



Outline

Bias



iLiNS_DOSE
iLiNS_DYAD_M
iLiNS_Zinc
JiVitA_3
JiVitA_4
LCNI_5
SAS_FoodSuppl
TanzaniaChild2
WASH_Bangladesh
WASH_Kenya
Bias
0.02
0.00
-0.02
AIPTW
CTMLE
CVTMLE
IPTW
TMLE

Software Use

- For estimating the various distributions in order to simulate, used SuperLearner, `sl3` package (https://github.com/tlverse/sl3).
- For non-CV TMLE and CV-TMLE estimators used the the `tmle3` package, part of the tlverse package (https://github.com/tlverse).
  - ✓ The original `tmle` (https://cran.r-project.org/web/packages/tmle/index.html) package works very well and also cross-validates the $\bar{Q}(A, W)$ part of the model, so adds robustness over estimating all components on the entire sample.
- For collaborative TMLE, or C-TMLE, used the excellent `ctmle` package (https://rdrr.io/cran/ctmle/).
  - ✓ Will not discuss much these results, because needs to be intelligently tuned and active area of research.
  - ✓ Has shown to be very effective at ameliorating the negative impacts of positivity, particularly when the TMLE-estimator is hopelessly biased.

MSE

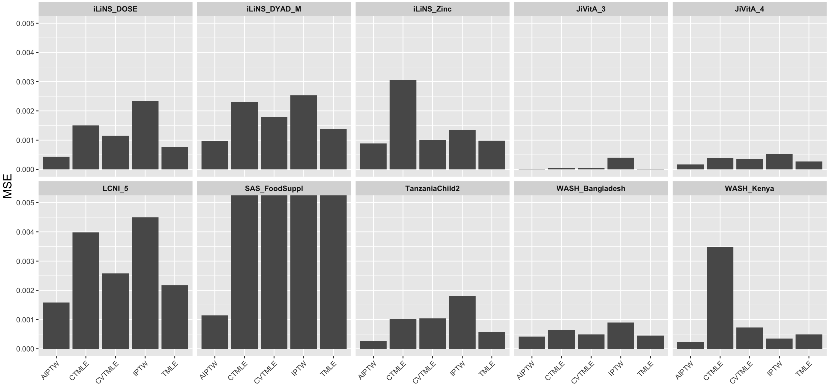


Figure 7: MSE across Ten Studies

95% CI Coverage

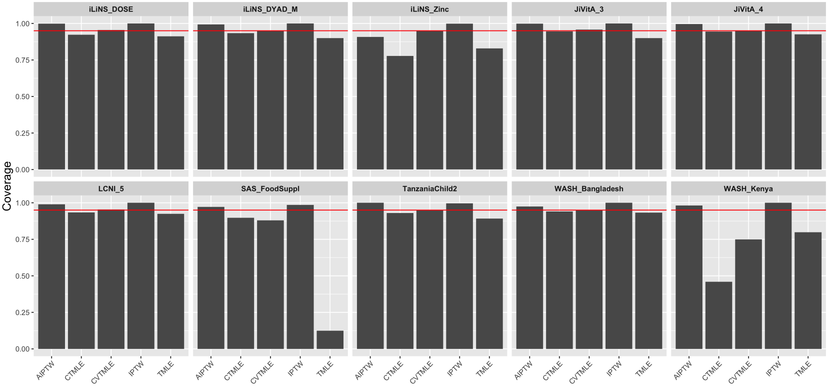
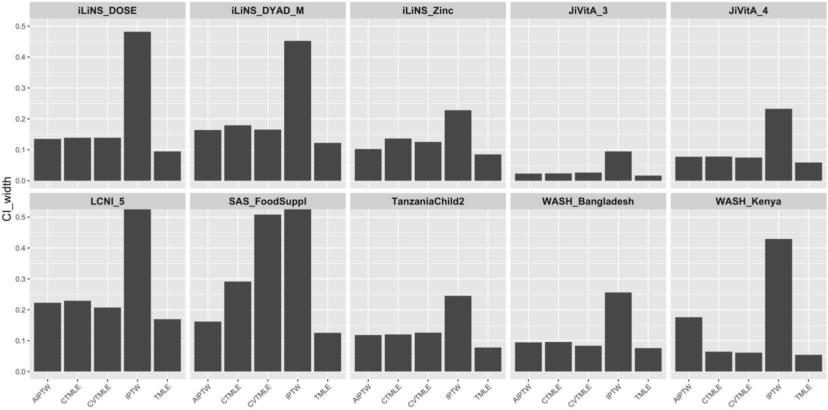


Figure 8: Coverage of Confidence Intervals in Ten Studies

95% CI Width

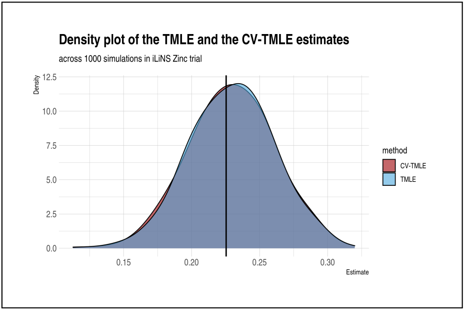


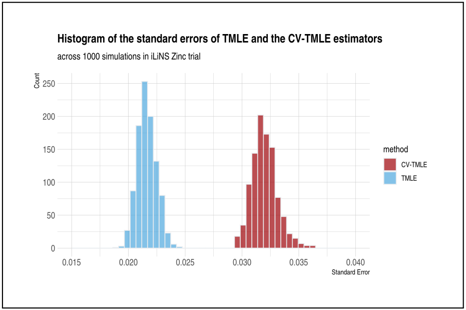
iLiNS_DOSE
iLiNS_DYAD_M
iLiNS_Zinc
JiVitA_3
JiVitA_4
LCNI_5
SAS_FoodSuppl
TanzaniaChild2
WASH_Bangladesh
WASH_Kenya
CI_width
0.0
0.1
0.2
0.3
0.4
0.5
AIPTW
CTMLE
CVTMLE
IPTW
TMLE

Summary thus far

- Considering both the efficiency of estimate and accuracy of inference, AIPTW and CV-TMLE both appear the most robust estimators.
- TMLE, IPTW perform well on average, but a few studies provide inaccurate inferences.
- In this case, the AIPTW tends to have overly conservative coverage (6/10 have greater than 99% coverage).
- However, width of CI's sometimes smaller, sometimes larger than CV-TMLE.

What happened with iLiNS Zinc study?

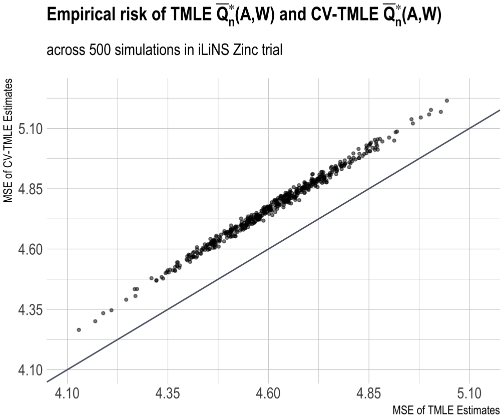




TMLE 95% CI Coverage = 0.82.

CV-TMLE 95% CI Coverage = 0.95.

Evidence of over-fitting



Compares the empirical MSE of the prediction model used in TMLE versus the unbiased cross-validated MSE of the prediction model used in CV-TMLE.

Outline

Modify the parameter a bit to make estimator more robust

- Petersen et al. (2012) in a paper about positivity, suggested use of realistic rules.
- These rules assign the desired treatment of interest (say $A = 1$), unless the propensity score is below some threshold (say $g(W) \equiv P(A = 1|W) < \alpha$).
- Usually, $g(W)$ unknown, so have to estimate, making this a data-adaptive parameter with parameter of interest

  $$\Psi(Q_0) = E_{W,0}\{E_0(Y \mid A = d(1, W), W) - E_0(Y \mid A = d(0, W), W)\}$$

  where $d(a, W) = a$ if $\hat{g}(W) > \alpha$ other $= 1 - a$.
- Reduces finite sample empirical positivity violations.

Results of Using Realistic Rule for two studies with poor performance by CV-TMLE

Table 4: TMLE/CV-TMLE Performance(without Realistic Rule)

Method	Study_ID	True_ATE	Variance	Bias	MSE	Coverage	CI_width
TMLE	SAS_FoodSuppl	-0.01516	0.08538	-0.22524	0.13612	0.124	0.12523
CV-TMLE	SAS_FoodSuppl	-0.01516	0.02267	-0.00679	0.02272	0.880	0.50820
TMLE	WASH_Kenya	-0.00387	0.00049	-0.00211	0.00049	0.798	0.05440
CV-TMLE	WASH_Kenya	-0.00387	0.00073	0.00065	0.00073	0.750	0.06138

Table 5: TMLE/CV-TMLE Performance under Realistic Rule

Method	Study_ID	True_ATE	Variance	Bias	MSE	Coverage	CI_width
TMLE	SAS_FoodSuppl	-0.00821	0.00884	-0.00504	0.00887	0.442	0.10001
CV-TMLE	SAS_FoodSuppl	-0.00821	0.01034	-0.02303	0.01087	0.871	0.32857
TMLE	WASH_Kenya	-0.00252	0.00019	0.00008	0.00019	0.939	0.05165
CV-TMLE	WASH_Kenya	-0.00252	0.00020	-0.00007	0.00020	0.956	0.05649

CV-TMLE Results

- SAS_FoodSuppl: coverage stays the same but CI gets smaller.
- WASH_Kenya: coverage is repaired to 95% and CI gets (a bit) smaller.

Outline

Take Aways

- Generally a good idea if making a estimation machine to avoid over-fitting by using a CV-approach (either for TMLE, or as suggested later by Chernozhukov et al. (2017) for AIPTW estimator double machine learning).
- Otherwise, it's complicated - differing goals (efficient estimates, proper coverage), and maybe get better performance by choosing different parameter, or doing both.

References

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, and Whitney Newey. Double/debiased/neyman machine learning of treatment effects. *American Economic Review*, 107(5):261–65, 2017.

Andrew Mertens, Jade Benjamin-Chung, John M Colford, Jeremy Coyle, Mark van der Laan, Alan E Hubbard, Sonali Dayal, Ivana Malenica, Nima Hejazi, Oleg Sofrygin, et al. Causes and consequences of child growth failure in low-and middle-income countries. *medRxiv*, 2020. URL https://www.medrxiv.org/content/10.1101/2020.06.09.20127100v1.

Maya L. Petersen, Kristin E. Porter, Susan Gruber, Yue Wang, and Mark J. van der Laan. Diagnosing and responding to violations in the positivity assumption. *Statistical Methods in Medical Research*, 21(1):31–54, 2012. ISSN 0962-2802. doi: 10.1177/0962280210386207.

M. van der Laan and S. Rose. *Targeted learning: causal inference for observational and experimental data*. Springer, 2011.

Mark J van der Laan, Eric C Polley, and Alan E Hubbard. Super learning. *Stat Appl Genet Mol Biol*, 6:Article25, 2007. doi: 10.2202/1544-6115.1309.

Wenjing Zheng and Mark J Van Der Laan. Asymptotic theory for cross-validated targeted maximum likelihood estimation. *U.C. Berkeley Division of Biostatistics Working Paper Series.*, (273), 2010. URL https://biostats.bepress.com/ucbbiostat/paper273.